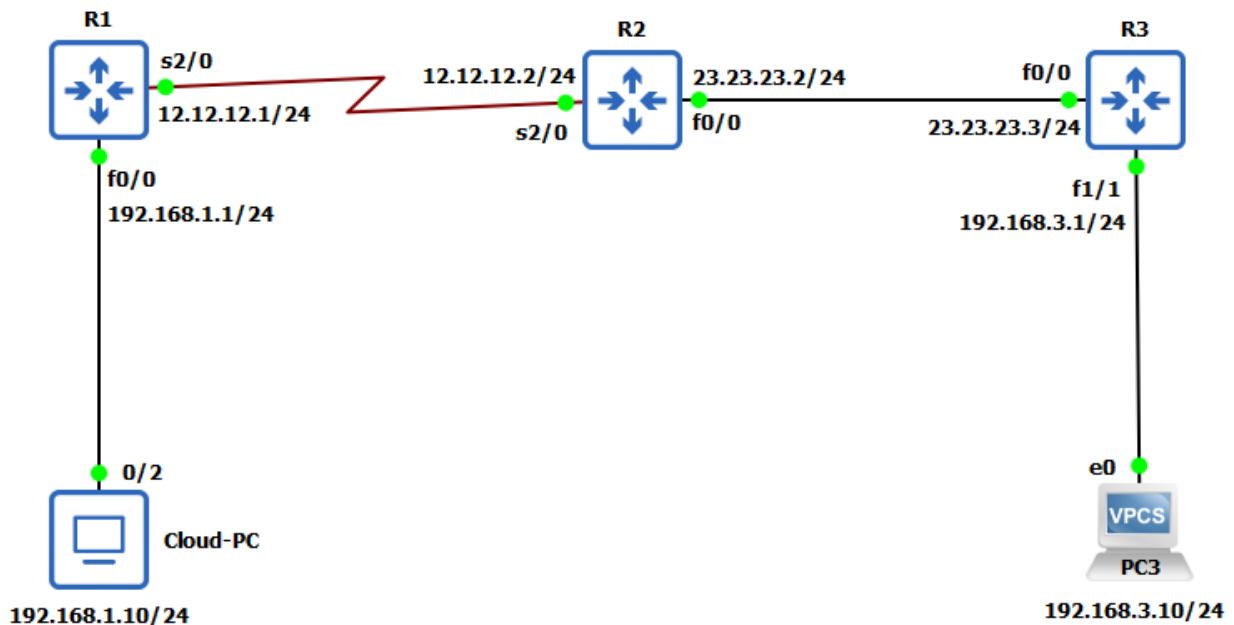


Default Routing



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna** and device access password as **cisco** (Refer Lab-1)
2. Configure static routes on R2 for R1's & R2's LAN network, assuming R2 as ISP router.
3. Configure default route on R1 & R3.
4. Cloud-PC & PC-3 should ping each other.

Note: Cloud-PC is a cloud which bridge the R1 router with loopback adapter on real PC.

Static default routing on R1 Router

```

R1#config t
R1(config)#ip route 0.0.0.0 0.0.0.0 12.12.12.2
R1(config)#exit
R1#
  
```

Static routing on R2 Router

```

R2#config t
R2(config)#ip route 192.168.1.0 255.255.255.0 12.12.12.1
R2(config)#ip route 192.168.3.0 255.255.255.0 23.23.23.3
R2(config)#exit
R2#
  
```

Static default routing on R3 Router

```
R3#config t
R3(config)#ip route 0.0.0.0 0.0.0.0 23.23.23.2
R3(config)#exit
R3#
```

✓ Verification & Testing:**R1#sh ip route static**

Codes: L - local, C - connected, **S - static**, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, *** - candidate default**, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is 12.12.12.2 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 12.12.12.2

R1#

R2#sh ip route static

Codes: L - local, C - connected, **S - static**, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

S 192.168.1.0/24 [1/0] via 12.12.12.1

S 192.168.3.0/24 [1/0] via 23.23.23.3

R2#

R3#sh ip route static

Codes: L - local, C - connected, **S - static**, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, *** - candidate default**, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is 23.23.23.2 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 23.23.23.2

R3#

R1#sh ip route 192.168.3.10

% Network not in table

Ping PC3 from R1

R1#ping 192.168.3.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 72/92/152 ms

R1#traceroute 192.168.3.10

Type escape sequence to abort.

Tracing the route to 192.168.3.10

VRF info: (vrf in name/id, vrf out name/id)

1 12.12.12.2 80 msec 64 msec 28 msec

2 23.23.23.3 36 msec 40 msec 40 msec

3 192.168.3.10 36 msec 52 msec 48 msec

R3#sh ip route 192.168.1.10

% Network not in table

Ping Cloud-PC from R3

R3#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 80/83/92 ms

R3#traceroute 192.168.1.10

Type escape sequence to abort.

Tracing the route to 192.168.1.10

VRF info: (vrf in name/id, vrf out name/id)

1 23.23.23.2 56 msec 56 msec 52 msec

2 12.12.12.1 84 msec 80 msec 80 msec

3 * * *

4 * * *

5 192.168.1.10 96 msec 108 msec 120 msec

Ping PC3 from Cloud-PC

C:\Users\Ratan>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time=3051ms TTL=61

Reply from 192.168.3.10: bytes=32 time=61ms TTL=61

Reply from 192.168.3.10: bytes=32 time=53ms TTL=61

Reply from 192.168.3.10: bytes=32 time=54ms TTL=61

Ping statistics for 192.168.3.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 53ms, Maximum = 3051ms, Average = 804ms

Ping Cloud-PC from PC3

PC3> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=125 time=63.281 ms

84 bytes from 192.168.1.10 icmp_seq=2 ttl=125 time=58.138 ms

84 bytes from 192.168.1.10 icmp_seq=3 ttl=125 time=65.138 ms

84 bytes from 192.168.1.10 icmp_seq=4 ttl=125 time=57.470 ms

84 bytes from 192.168.1.10 icmp_seq=5 ttl=125 time=66.392 ms

Ratan

Important Commands:

```
sh ip route  
sh ip route static  
sh ip route static | begin Gateway
```

Ratan